

MilkWise — iOS & Android Build Guide

Project: MilkWise React Native App

Repo: [koenswings/milkwise-rn](#) · Local: [varia/milkwise/](#)

Framework: Expo SDK 54 / React Native 0.81

Build system: EAS Build (Expo Application Services — cloud builds)

Date: 2026-09-01 (updated)

Overview

MilkWise is a React Native app built with Expo. Development and testing run locally via Expo Go. Production builds for the App Store (iOS) and Google Play (Android) are compiled in the cloud via EAS Build — **not** on the Pi, because the Pi's ARM architecture cannot run the Hermes JavaScript compiler required for store submissions.

Prerequisites

Accounts

Service	Purpose	Cost
Expo account	EAS Build, project hosting	Free
Apple Developer Program	iOS App Store submission	\$99/year
Google Play Console	Android Play Store submission	\$25 one-time

Local tools

```
# Node.js (already installed)
node --version    # 22.x

# Expo CLI
npm install -g expo-cli eas-cli

# Confirm EAS login
eas whoami
```

App identifiers (already configured in `app.json`)

Platform	Identifier
iOS	<code>com.koenswings.milkwise</code>
Android	(auto from slug <code>milkwise</code>)
Expo project ID	<code>16e4e7d9-35a5-4604-9046-bf630253ab73</code>

Development (Expo Go)

For day-to-day development, no build is needed. The app runs inside the Expo Go client on a physical device or simulator.

1. Configure the API URL

Create `.env.local` in `varia/milkwise/` (gitignored):

```
EXPO_PUBLIC_API_URL=http://100.85.108.118:3333
```

This points to the MilkWise web app running on idea02 (Tailscale IP). Swap for a local address if running the web app locally.

2. Start the Metro bundler

```
cd varia/milkwise
npx expo start --lan --port 8082
```

Open the QR code in Expo Go on your device (must be on the same Wi-Fi network as the Pi, or connected via Tailscale).

3. TypeScript check

```
npx tsc --noEmit
```

Production Builds (EAS Build)

EAS Build runs in the cloud on Expo's infrastructure. You push your source code; Expo compiles it and returns a signed `.ipa` (iOS) or `.aab` (Android).

EAS configuration (`eas.json`)

```
{
  "cli": {
    "version": ">= 12.0.0",
    "appVersionSource": "local"
  },
  "build": {
    "development": {
      "developmentClient": true,
      "distribution": "internal"
    },
    "preview": {
      "distribution": "store"
    },
    "production": {
      "distribution": "store",
      "autoIncrement": true
    }
  },
  "submit": {
    "production": {
      "ios": {
        "ascApiKeyPath": "./AuthKey_HYRJUKYMTB.p8",
        "ascApiKeyId": "HYRJUKYMTB",
        "ascApiKeyIssuerId": "a0b17ef9-b9bc-4c4d-9b27-074fe6587570",
        "ascAppId": "6796410350"
      }
    }
  }
}
```

Three profiles are defined:

Profile	Purpose	Distribution
development	Dev client with debugging	Internal (TestFlight / internal track)
preview	Near-production test build	Internal
production	Store submission build	App Store / Google Play

iOS Build

One-time setup (completed 2026-09-01)

Step	Status	Notes
Apple Developer account	Enrolled	<code>koen.swings@me.com</code>
Bundle ID registered	Done	<code>com.koenswings.milkwise</code>
EAS credentials (distribution cert + provisioning profile)	Done	Managed by EAS
App Store Connect app record	Done	App ID <code>6796410350</code>
Apple API key (.p8)	Done	Key ID <code>HYRJUKYMTB</code> , Issuer <code>a0b17ef9-b9bc-4c4d-9b27-074fe6587570</code> , file at <code>varia/milkwise/AuthKey_HYRJUKYMTB.p8</code>
First preview build	Done	<code>.ipa</code> artifact produced by EAS on 2026-09-01
TestFlight submission	In progress	<code>eas submit --platform ios --latest</code>

To re-run credentials setup (e.g. after cert expiry):

```
cd varia/milkwise
EXPO_TOKEN=gDC-sqF9kLgF-k4jfuPzWdQRE-AuUoyBWpFqwItQ
EXPO_APPLE_ID=koen.swings@me.com eas credentials --platform ios
```

Select **"Set up all the required credentials to build your project"** → EAS manages everything automatically. Skip push notifications (not yet wired up).

Build

```
cd varia/milkwise

# Preview build (for TestFlight testing)
```

```
eas build --platform ios --profile preview

# Production build (for App Store submission)
eas build --platform ios --profile production
```

EAS prints a build URL. The build runs in the cloud (10–20 minutes). When complete, download the [.ipa](#) or submit directly.

Submit to App Store

```
eas submit --platform ios --latest
```

This submits the most recent iOS build to App Store Connect. Review in App Store Connect, fill in metadata, and submit for Apple review.

Android Build

One-time setup

1. **Google Play Console** — create an account at play.google.com/console. Pay the \$25 registration fee.
2. **Keystore** — EAS manages the Android signing keystore automatically. Run once:

```
eas credentials --platform android
```

Select "Let EAS manage credentials". The keystore is stored securely on Expo's servers.

Build

```
cd varia/milkwise

# Preview build (APK — can install directly on device)
eas build --platform android --profile preview

# Production build (AAB — for Play Store)
eas build --platform android --profile production
```

The production build produces an `.aab` (Android App Bundle), which the Play Store requires.

Submit to Google Play

```
eas submit --platform android --latest
```

Or manually upload the `.aab` in the Google Play Console under **Production → Create new release**.

Both Platforms Simultaneously

```
eas build --platform all --profile production
```

Version Management

The app version is defined in `app.json`:

```
{
  "version": "1.0.0"
}
```

With `"autoIncrement": true` in the `production` profile, EAS automatically increments the build number (`versionCode` on Android, `buildNumber` on iOS) with each production build. The user-facing version string (`1.0.0`) must be bumped manually in `app.json` before a new release.

Shared Core — Sync Rule

Two files are shared between `milkwise-rn` and `baby-milk-tracker` (the web app). Any change to either must be applied to **both repos in the same session**:

File	Purpose
<code>src/types/index.ts</code>	Shared type definitions

File	Purpose
<code>src/lib/calculations.ts</code>	Business logic (smoothed formula, stats, predictors)

Environment Variables

Variable	Where	Purpose
<code>EXPO_PUBLIC_API_URL</code>	<code>.env.local</code> (gitignored)	Web app API base URL

For production builds, the API URL should point to the production MilkWise instance (idea02 or future hosting). Update `.env.local` or set via EAS environment variables before building.

Known Limitations

Issue	Status
Push notifications	Skipped during credentials setup — not yet wired up; remove capability from <code>app.json</code> before App Store submission or wire up APNs
ARM (Pi) cannot build locally	Use EAS cloud builds for any store submission
Expo Go requires SDK 54-compatible version	Install from App Store / Play Store
Home screen widget	Planned for v1.1
Supabase household sync	Designed, not implemented (v1.1)

Quick Reference

```
# Login via token (preferred on Pi – avoids interactive prompts)
EXPO_TOKEN=<token> eas whoami

# Check current project
EXPO_TOKEN=<token> eas project:info

# iOS preview
EXPO_TOKEN=<token> eas build --platform ios --profile preview
```

```
# Android preview  
EXPO_TOKEN=<token> eas build --platform android --profile preview
```

```
# Both production  
EXPO_TOKEN=<token> eas build --platform all --profile production
```

```
# Submit iOS (uses .p8 key configured in eas.json submit block)  
EXPO_TOKEN=<token> eas submit --platform ios --latest
```

```
# Submit Android  
EXPO_TOKEN=<token> eas submit --platform android --latest
```